MIL-P-15736/4(SHIPS)
9 August 1965
SUPERSEDING
MIL-P-19421A(SHIPS)
6 July 1956
(See 6.2)

MILITARY SPECIFICATION

POWER SUPPLY, METALLIC RECTIFIER, MISSILE TESTING

1. SCOPE

1.1 This specification covers missile testing metallic rectifier power supplies for Naval shipboard use in servicing special aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-2212 - Controllers, Alternating-Current, Naval Shipboard.
MIL-S-3950 - Switches, Toggle.
MIL-M-10304 - Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for.
MIL-P-15736 - Power Supplies, Metallic Rectifier, Naval Shipboard.

DRAWINGS

BUREAU OF SHIPS
9000-S6202-73882 - Plug and Receptacle.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. - The power supply shall be in accordance with class R of MIL-P-15736 and as specified herein. (Where any conflict exists between MIL-P-15736 and this specification, the latter shall govern.)

3.2 Electrical ratings. - Electrical ratings shall be as specified in table I.

Table I - Electrical ratings.

Parameters	Requirements
A.c. volts	440 ± 5 percent
Frequency	60 cps ± 5 percent
Phase	3
D.c. volts (output)	26 to 40 volts (see 3.2.2)
Type of load	Resistive, inductive, capacitive and battery charging, Nickel-Cadmium

3.2.1 Operation. - The power supply shall be operable under any of the specified conditions of output voltage and load either with or without automatic current limiting control. Operation with or without current

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limiting shall be by operation of a selector switch (see 3.2.2.2.2). Internal adjustments shall be provided for adjusting the specified current limiting valves to compensate for component design tolerance and aging.

3.2.2 Direct current (d.c.) output (special performance requirements). -

3.2.2.1 Voltage regulation and adjustment. -

3.2.2.1.1 Voltage regulation. - The output voltage from 26 to 40 volts shall be automatically regulated to within plus or minus 1 percent of the selected voltage from no load to 50 amperes. From 50 to 150 amperes load, the voltage regulation shall be within plus 1 volt, minus 3 volts of the no load selected voltage from 26 to 40 volts, with specified valuation of input voltage, frequency and room ambient. Voltages shall be regulated by the power supply at a load point remote from the power supply. The load voltage drop from the power supply to the load may be any value from 0 to 4 volts at 50 amperes load. This feature shall be described as remote voltage regulation and may be accomplished by one or both of the following means:

(a) By using 2 wire remote voltage sensing leads.

(b) By voltage compounding through use of a load sensitive control (d.c. measuring shunt or transductor).

If remote voltage sensing leads are used the control shall be such that the signal current in the sensing leads does not exceed 1 ampere under all specified operating conditions.

3.2.2.1.2 Voltage adjustment. - The power supply shall provide continuous output voltage adjustment of 26 to 40 volts d.c. measured at the load. Adjustment rheostat shall provide resolution of adjustment no greater than 0.25 volts. Either local or remote adjustment of the output voltage shall be possible by operation of the "local-remote" selector switch to the desired position (see 3.6.2). When set at "local" position, voltage adjustment shall be by operation of the local rheostat. When set at "remote" position, the local voltage adjusting rheostat shall be circuited to have no control of the output voltage. Remote voltage adjustment shall be by operation of a remote voltage adjustment rheostat connected through corresponding load, voltage sensing, and control terminals of the power supply. The remote rheostat shall not normally be furnished with the power supply.

3.2.2.2 Current and selective limiting. -

3.2.2.2.1 Current limiting. - Current limiting is required to prevent overloading the power supply without interruption of power under certain conditions of operation. For example, under battery charging, selective limiting is required to prevent over-heating of connections at the battery terminals as well as to protect the power supply against excessive currents.

3.2.2.2.2 Selective limiting. - Current limiting shall be provided by automatic control to limit the load current to 50 amperes ± 10 percent under conditions of load equivalent to a 100 ampere hours nickel-cadmium battery at initial complete discharge. Transient current surge under current limiting shall recover to hold 50 amperes ± 10 percent within 2 seconds from the time of initial application of load. During the current limiting period the output voltage may drop below the preset value providing the drop in output voltage is not great enough to limit the current below 45 amperes. As the load impedance increases to cause the load current to drop below the nominal limiting value of 50 amperes ± 10 percent, the output voltage shall then automatically be regulated to the selected voltage setting as specified herein (see 3.2.2.1.1). The use of any series resistor in the load circuit to limit the current by direct action of its own impedance is not permitted. The current limiting provision specified herein is required primarily for operation under conditions of charging nickel-cadmium batteries. Current limiting shall be effective to operate as specified above when charging nickel-cadmium batteries from 25 to 100 ampere hours. Operation of the power supply with or without the above current limiting shall be optional by operation of a selector switch (see 3.7.3). With the selector switch in "on" position, the above current limiting shall function, with the selector switch in "off" position, the above current limiting shall not function and the power supply shall continuously regulate on voltage as specified herein.

3.2.2.3 Transient voltage regulation performance and adjustment. -

3.2.2.3.1 Transient voltage regulation performance. - With a load resistance of 0.2 ohm applied at the output terminals of the power supply (150 amperes at 30 volts), the output voltage as measured at the power

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supply shall not fall below 66 percent of the no load setting for all settings from 26 to 28 volts inclusive and not below 62 percent of the no load setting for all settings from 28 to 32.5 volts. Voltage recovery under these conditions of load shall be such that for no load settings of 26 volts and above the voltage shall not remain below 20 volts for more than 13 milliseconds and shall recover to hold within plus 1 volt, minus 3 volts of initial no load setting within 0.1 second. Following sudden change from 50 amperes load to no load the voltage rise shall not exceed 25 percent of the initial no load setting for all settings from 26 to 40 volts.

3.2.2.3.2 Adjustment of transient voltage regulation performance.- The transient voltage regulation performance required for proper operation of loads to which the power supply will be connected is described in 3.2.2.3.1. In many cases, the load will consist of a rotary inverter whose input circuit is controlled by devices with transient performance similar to that of the power supply. In order to minimize the possibility of voltage oscillation, the power supply shall provide a control device for manually adjusting the response time (time for recovery of voltage following load application or change (see 3.7.3(c)). This device shall be known as the "stability control". At one extreme of adjustment of this control, the transient performance of the power supply shall conform to 3.2.2.3.1. As the stability control is adjusted, the response time of the power supply shall progressively and smoothly increase to a total response time of not less than 0.5 second. Optionally the stability control may consist of a selector switch with a minimum of 3 positions, representing progressively increasing response time.

3.2.2.4 Ripple voltage.- The ripple content of the output voltage shall not exceed 1 percent as an r m s measurement and shall not exceed 3 percent as a peak-to-peak measurement. This ripple requirement shall apply to any load from 0 to 75 amperes.

3.2.3 Remote sensing.- The d.c. output of the power supplies shall terminate in two four-pole receptacles, in accordance with symbol 758 of Drawing 9000-S6202-73882. Two of the poles shall be used for connecting remote sensing leads and two for connecting d.c. power leads ("A" terminal is for positive power lead, "C" terminal is for negative power lead, "B" terminal is for positive sensing lead, "N" terminal is for negative sensing lead). All receptacles shall be equipped with the corresponding male plug assembly. Provision shall also be made for making permanent connections within the power supply by including a terminal board for convenient connections internally to load sensing and remote control (see 3.2.2.1.2) terminals.

3.2.4 Duty.- The power supply shall be rated to give continuous duty at 50 amperes load and intermittent duty of 200 amperes from start to 15 seconds, 150 amperes from 15 seconds to 30 seconds, 135 amperes from 30 seconds to 1 minute, 120 amperes from 1 minute to 2 minutes, 105 amperes from 2 minutes to 3 minutes, 85 amperes from 3 minutes to 4 minutes, and 75 amperes from 4 to 15 minutes. The above intermittent duty cycle shall be considered as starting with the power supply at no load and with the equipment (including all components) in an ambient of 50°C. such that all components are warmed to a temperature at least as high as the surrounding ambient.

3.3 Enclosure.- The enclosure shall be dripproof protected with a hinged front access door. All meters shall be mounted on the door. Unless otherwise specified herein, all switches and miscellaneous controls shall be mounted on the door for external operation. The output receptacles shall be mounted on the front, above the door.

3.4 Mounting.- The equipment shall be designed for deck or bulkhead mounting.

3.5 Size and Weight.- The maximum size and weight of the equipment shall be as follows:

Width	22 inches
Height	30 inches
Depth	18 inches
Weight	320 pounds

3.6 Component parts.-

3.6.1 Meters.- A voltmeter and ammeter shall be provided for measuring the total output current and output voltage. Meters shall be ruggedized (HI shockproof), 3-1/2-inch round panel type in accordance with MIL-M-10304. The voltmeter shall be with 50 volt full scale range (0 to 50 volts). The ammeter shall be with 80 amperes full scale range (0 to 80 amperes). The ammeter shall operate from a 50 millivolt external

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shunt. All meters and shunts shall be furnished integral with the power supply. Terminals shall also be provided for connection of a remote indicating voltmeter and ammeter.

3.6.2 Switches. - The following switching means shall be provided:

(a) A three-pole manual LVRE controller shall be provided for on-off operation at the a.c. input circuit. The controller shall provide overload protection to the power supply shall be flush mounted, and shall be in accordance with MIL-C-2212.

(b) An on-off switch shall be provided for operation with or without the current limiting control feature (see 3.2.2.2.2).

(c) A "local-remote" selector switch shall be provided to give selection of either local or remote voltage adjustment (see 3.2.2.1.2).

(d) An ammeter "press-to-read" switch shall be provided. The switch shall consist of a momentary contact toggle switch in accordance with MIL-S-3950. Normal position of switch contacts shall disconnect the ammeter from the load circuit to require operation of the switch to read the load current. The switch shall be spring return to normal position.

3.7.3 Miscellaneous controls. - The following miscellaneous controls shall be provided:

(a) A rheostat shall be provided for continuous adjustment of the output voltage over the specified range of 26 to 40 volts.

(b) An indicator light shall be provided and connected to indicate "power on" in the a.c. input circuit.

(c) Switch or rheostat or both as required for adjustment of power supply transient voltage regulation characteristics (see 3.2.2.3.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-P-15736 and shall include the tests specified in 4.2.1 through 4.2.4 herein for selective, periodic and qualification inspection.

4.2 Tests. -

4.2.1 D.c. output and special performance. - All tests shall be performed under operation with remote voltage regulation simulating approximately 4 volts load voltage drop from power supply to the load at 50 amperes load.

4.2.2 Transient voltage regulation performance. - Tests on transient voltage regulation performance shall be made using a resistance load equal to 0.2 ohm (150 amperes at 30 volts). The load shall be applied directly at the output terminals of the power supply under each condition of input voltage, frequency and ambient temperature as specified in MIL-P-15736. The range of no-load output voltage settings shall be covered from 26 to 32.5 volts in steps not exceeding 2 volts under each of the specified 10 conditions of input voltage, frequency, and ambient temperature. Application and interruption of the load shall be made by operation of a magnetic contactor connected with the main contacts in series with the load. Recordings shall be made of both the output voltage and current with sufficient scale of amplitude and time to give good readability.

4.2.3 Selective limiting. - Test on selective current limiting shall be made using a load equivalent to nickel-cadmium batteries connected in parallel at complete discharge. The load shall be applied directly at the output terminals of the power supply under the conditions of rated input voltage, frequency and room ambient temperature. Test runs shall be made at a voltage setting corresponding to the full charge rated battery voltage (if batteries are used) or at both initial voltage settings of 26 and 40 volts (if other than battery loads are used). Within 5 seconds after application of load the load impedance shall be gradually increased until the load current has dropped to 5 amperes. The total time for varying the load impedance to reach 5 amperes shall be not less than 10 minutes. Recordings shall be made simultaneously of both the output voltage and current with sufficient scale of amplitude and time to give good readability. From results of this test it shall be determined if requirements of 3.2.2.2.2 are satisfied.

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4.2.4 Inductive and battery loads. - Output load tests shall be performed to determined satisfactory performance on inductive loads. At least one voltage regulation run shall be made using an inductive (d.c. motor) load having an inrush current of 125 to 150 amperes and one using a battery type load. The battery type load shall be equivalent to a 50 ampere-hour nickel-cadmium battery initially at complete discharge condition. These tests may be made at rated input voltage and frequency and room temperature. Any indication of unsatisfactory or questionable performance with other than resistive loads shall necessitate additional load performance tests as outlined herein. The extent of additional testing shall only consist of that necessary to determine conformance or nonconformance to the requirements of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-P-15736.

6. NOTES

6.1 The notes specified in MIL-P-15736 are applicable to this specification.

6.2 Superseding information. - This specification supersedes MIL-P-19421A. The power supply described by MIL-P-19421A was identified as PP1458/U, type I.

Preparing activity:
Navy - SH
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